



Assignment 1

Performance Analysis of Different Edge Detection Techniques

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MENG - Electrical and Computer Engineering

Course Code: ELEC-8280: Image Processing

Date: Feb. 1, 2026

Summary

In this assignment, we analyze the performance analysis of different edge detection techniques (Sobel, Prewitt, Laplacian of Gaussian, and Canny edge detectors) on the famous Lena image. To be able to quantitatively analyze the performance we will add Gaussian noise to the image at five different variance levels (var = 0.01, 0.05, 0.1, 0.5, 1), and then we use Root Mean Square Error (RMSE) to compare the results of each edge detector.

Equation used to calculate the RMSE:

$$RMSE = \sqrt{\frac{1}{R * C} \sum_{i=1}^R \sum_{j=1}^C |f(i,j) - f(\acute{i},j)|^2}$$

R: number of rows
C: number of columns
 $f(i,j)$: reference image
 $f(\acute{i},j)$: Image with detected edges using different edge detection methods with different variance levels of gaussian noise

Reference image is the image with edges using a Canny Edge detector (threshold = 0.1, sigma = 1) and no Gaussian noise.

Then, we analyzed the performance of the Canny Edge detector (in terms of detected edges) with different threshold values and explained the results.

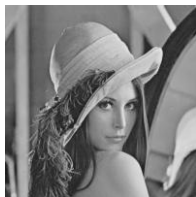
After that, we discussed the advantages of Canny edge detector, basic criteria for edge detection and how Canny edge detector can satisfy and why it is better than other edge detectors, how well Canny edge detectors preform vs other edge detectors.

Finally, we discussed directional edge detectors, what in unconstrained natural environment, what is directional edge detectors and how does a directional edge detector preform with image with unconstrained natural environment and why it is better.

Analysis and Results

Q1

After converting Lena image to greyscale and manually adding five different variance levels (var=0.01, 0.05, 0.1, 0.5, 1) of Gaussian noise with all means =0.



Original grey img.



Var.=0.01



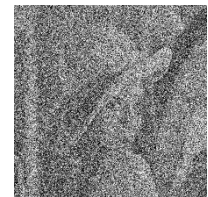
Var.=0.05



Var.=0.1



Var.=0.5



Var. =1

Also, using a Canny Edge detector (threshold = 0.1, sigma = 1) when there is no Gaussian noise to analyze the performance of other edge detectors, as shown in figure 1.



Figure 1: Reference image using Canny edge detector

We obtain the following results (Quantitatively analyzation of the performance):

RMSE value of different Edge Detectors with different variance of noise				
Variance	Sobel	Prewitt	LoG	Canny
0.01	0.30907	0.30867	0.38810	0.56638
0.05	0.33050	0.33048	0.46055	0.61356
0.1	0.33378	0.33301	0.48008	0.62587
0.5	0.33018	0.33027	0.50086	0.63973
1	0.32789	0.32765	0.50373	0.64183

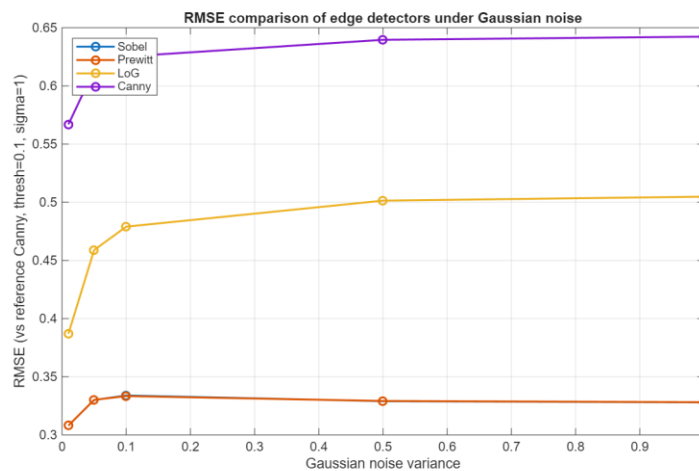
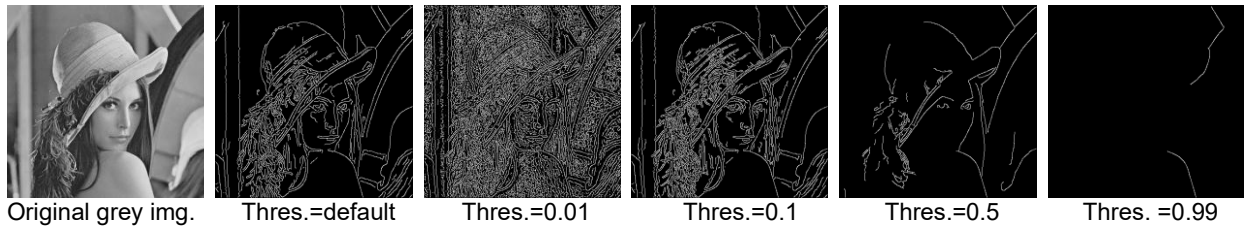


Figure 2: graph representation of the RMSE results of different edge detectors

Interpretation of the results: The RMSE values reflect how different each edge detection algorithm is. From the graphs, we can see the RMSE value increases as the variance value increases from 0.01 to 1 for LoG and Canny, but for Sobel and Prewitt, it increases and then nearly stabilizes. As the noise increases, the detected edges begin to be very different from the reference image when utilizing the Canny and LoG methods. When observing the Prewitt and Sobel methods, they had a constant performance with a value of 0.33 throughout all variance levels. Because RMSE calculates pixel-to-pixel differences between the reference image and testing image, larger RMSE values reflect a loss of accuracy.

Q2

Analyzing the performance of Canny edge detector with varying threshold values (default, 0.01, 0.1, 0.5, 0.99)



Interpretation of the results: As it can be seen in the Canny edge detector results, the threshold is essentially deciding how "strong" a change in intensity must be to qualify as an edge. When the threshold is very low (0.01), the detector becomes too sensitive and therefore detects a lot of changes in edge intensity. This is why the edge map appears "crowded," because it contains a lot of texture and false edges. With the default thresholds, the results look good – they balance the important edges that are retained with the unwanted details that are rejected – although it should be noted that MATLAB chose these thresholds automatically, which is probably why they are fairly good. When the threshold is 0.1, the important edges are still well represented, but some details are missing. When the threshold rises to 0.5, only the strong edges remain, which means that the edge map is thin and a lot of real edges are missing because they are not very strong. When the threshold is increased to 0.99, almost nothing is accepted, which means that only a few strong edges remain and most of the important details are missing.

Q3

- 1) Briefly discuss the strength of Canny Edge detector over rest of detection schemes mentioned above.

Canny is better than Sobel, Prewitt, and LoG because it is not only an edge detector, but also an entire process. As Canny smooths and removes noise from the image, then continues with edge detection, and ends with edge refinement to make the edges thinner and more uniform.

Therefore, the edges detected by the Canny algorithm tend to look more realistic and uniform.

- 2) Advantages of Canny Edge Detector

- It gives thin edges, rather than thick, and blurry boundaries.
- It uses two thresholds (hysteresis), so it keeps weak edges only when they are part of a real boundary.
- It includes Gaussian smoothing, so it is less sensitive to Gaussian noise.

- It typically produces better outputs with fewer random edges, because the Gaussian smoothing step [controlled by σ] suppresses high frequency noise before computing the gradient.

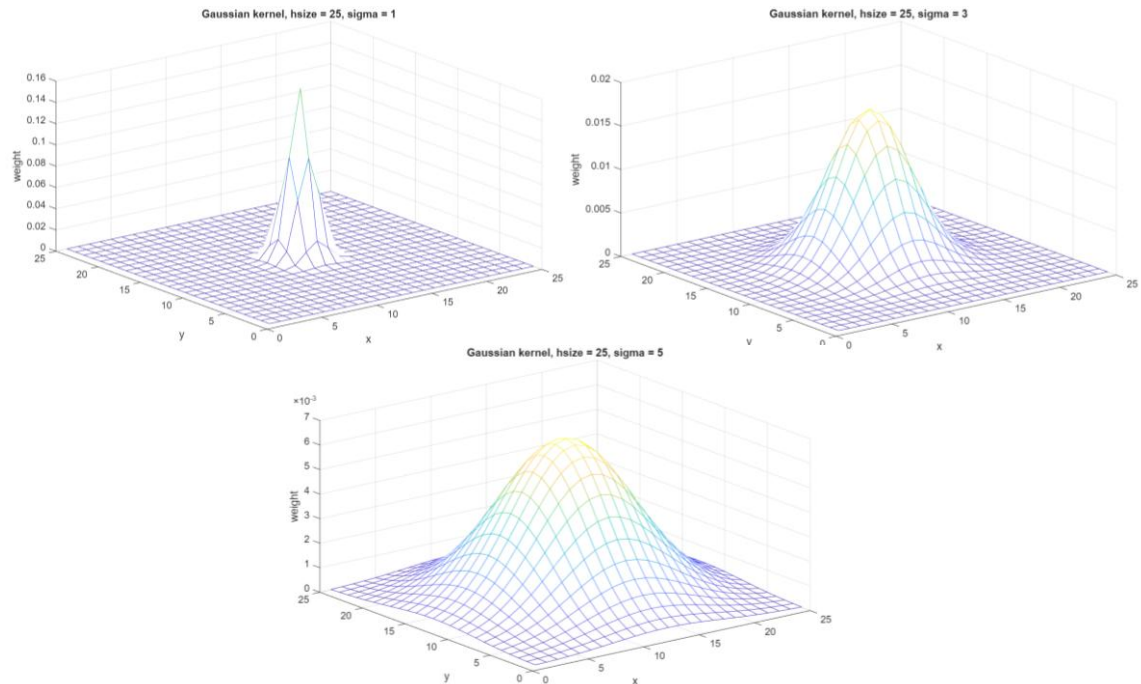


Figure 3: 3D mesh of Gaussian kernel (hsize = 25) for $\sigma = 1, 3, 5$

For larger σ means that edge detector will spread the weights more, meaning stronger smoothing, which reduces noise but also blurs fine details and slightly shifts edge location.

- You can control it using threshold and sigma.
- 3) Basic criteria for edge detection, how Canny edge detector can satisfy? Why are the other detectors do not satisfy?
 The overall objectives of edge detection are: to mark real edges, to mark them in the correct position, and to mark them without multiple edges for the same boundary. Canny meets these objectives because it first smooths the image, then it uses the gradient to mark the strength of edges, then it uses non-maximum suppression to mark only the strongest point along each edge so that edges are thin, and finally it uses hysteresis to mark weak edges only if they are connected to strong edges. The other detectors such as Sobel and Prewitt are simple gradient masks, and they don't respond that well to noise and texture. However, they may produce thicker edges or false edges, as they lack a good "decision step" like hysteresis. LoG is a second derivative, which looks for zero-crossings, and this may produce false edges and non-continuous edges depending on the amount of noise and smoothing applied. Generally, these methods do not control edge thickness, false edges, and continuity as well as the Canny method.
 - 4) How well does it perform in presence of noise compared to others?
 Gaussian noise is usually better handled by the Canny method because the Gaussian reduces the noise with a Gaussian smoothing and the hysteresis helps to eliminate the noise while maintaining the edges. Sobel and Prewitt detectors are likely to produce many false edges when the noise is higher. Similarly, the LoG method is also unstable in the presence of noise. Hence, in the presence of noise, the Canny method is the best and produces the cleanest edges.

Q4

1) What are directional edge detectors?

Edge detectors that react more strongly to edges in a particular direction, such as diagonal, vertical, or horizontal edges.

2) What do we mean by unconstrained natural environment?

The real-world scenes, which could be both outdoor and indoor, have uncontrolled lighting, shadows, textures, and many objects in the backgrounds.

3) What do we mean by “directional”?

This term refers to a detector that is designed to look in a certain direction and, therefore, is sensitive to variations in intensity in that direction.

4) How does a directional edge detector perform in such an environment?

The detector is useful in highlighting edges in a certain direction and suppressing many texture variations that are not in that direction, and therefore, a cleaner result is obtained.

5) Why do directional edge detectors perform better in images of unconstrained natural environment?

This is because in natural environments, there are many textures and variations in intensity, and therefore, a more effective separation of object boundaries can be achieved.